

VERHNYAYA TOM, Russian Federation

WMO: 314590

Lat: 51.350N

Lon: 130.433E

Elev: 261

StdP: 98.23

Time Zone: 9.00 (E09)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.3	(c) -40.3	(d) -44.8	(e) 0.0	(f) -42.0	(g) -42.9	(h) 0.1	(i) -40.1	(j) 6.4	(k) -13.6	(l) 5.6	(m) -15.0	(n) 0.0	(o) N/A	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	29.1	21.0	27.5	20.2	25.9	19.4	22.8	26.9	21.7	25.8	20.6	24.4	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.8	25.3	20.2	15.4	24.1	19.2	14.4	22.9	69.3	26.9	65.0	26.1	60.7	24.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.3	(b) 6.4	(c) 5.3	(e) -44.6	(f) 31.4	(g) 2.6	(h) 1.3	(i) -46.5	(j) 32.4	(k) -48.0	(l) 33.1	(m) -49.5	(n) 33.9	(o) -51.4	(p) 34.8		
(a) 8.3	(b) 6.4	(c) 5.3	(e) -44.1	(f) 24.9	(g) 2.1	(h) 1.2	(i) -45.7	(j) 25.7	(k) -46.9	(l) 26.4	(m) -48.1	(n) 27.0	(o) -49.6	(p) 27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.9	-28.7	-22.9	-12.5	0.9	9.1	15.7	18.6	16.3	9.3	-0.7	-15.5	-26.1		
	DBStd	17.39	4.88	5.57	6.20	4.63	4.26	3.08	2.75	2.91	4.20	5.32	6.67	5.49		
	HDD10.0	5445	1200	922	699	275	67	1	0	0	65	332	765	1120		
	HDD18.3	7832	1458	1155	957	523	286	89	30	77	273	590	1015	1378		
	CDD10.0	719	0	0	0	1	40	173	268	194	42	1	0	0		
	CDD18.3	64	0	0	0	0	1	11	39	13	1	0	0	0		
	CDH23.3	1017	0	0	0	0	54	271	532	151	9	0	0	0		
	CDH26.7	212	0	0	0	0	6	58	128	20	1	0	0	0		
(14) Wind	WSAvg	1.0	0.4	0.6	1.1	1.6	1.5	1.3	1.1	1.1	1.4	1.2	0.8	0.3		
(15) Precipitation	PrecAvg	764	11	10	23	44	79	103	148	136	92	62	36	20		
	PrecMax	1039	24	29	47	94	163	194	250	212	163	132	73	46		
	PrecMin	579	3	0	2	12	30	36	27	53	33	16	9	8		
	PrecStd	108	6	7	13	21	33	36	49	40	32	24	15	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.1	-2.8	9.9	19.7	27.5	30.6	31.6	28.9	24.6	16.9	5.0	-6.2		
		MCWB	-11.3	-5.0	3.2	10.1	15.3	20.6	22.3	22.7	17.1	10.5	1.4	-7.3		
	2%	DB	-12.9	-6.2	5.1	15.9	25.1	28.6	29.6	26.6	21.9	14.0	0.0	-9.9		
		MCWB	-13.7	-8.2	0.4	7.8	13.9	19.8	21.5	20.8	15.7	7.7	-2.0	-10.6		
	5%	DB	-15.7	-9.0	2.1	13.2	22.4	26.7	28.0	24.9	19.8	11.3	-2.2	-13.1		
		MCWB	-16.3	-10.6	-1.7	6.0	13.1	18.7	21.1	20.1	15.2	6.1	-4.0	-13.8		
	10%	DB	-18.4	-11.6	-0.4	10.7	19.8	24.7	26.5	23.5	17.8	8.7	-4.3	-16.1		
		MCWB	-18.9	-12.8	-3.4	4.7	11.8	17.7	20.8	19.4	14.0	4.8	-5.7	-16.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.1	-4.9	4.1	10.5	16.4	23.3	24.8	24.1	18.9	11.3	1.1	-7.4		
		MCDB	-10.3	-2.7	9.7	18.8	23.6	28.5	28.6	27.0	21.7	15.5	4.7	-6.5		
	2%	WB	-13.6	-8.1	0.9	8.1	15.2	21.1	23.3	22.1	17.3	8.6	-1.8	-10.6		
		MCDB	-12.9	-6.4	4.2	14.0	22.8	26.6	27.4	25.2	20.3	12.4	-0.2	-9.9		
	5%	WB	-16.3	-10.6	-1.4	6.5	13.9	19.8	22.3	20.8	16.0	6.9	-3.9	-13.8		
		MCDB	-15.7	-9.2	1.9	12.1	21.3	24.9	26.6	23.6	18.6	10.6	-2.3	-13.1		
	10%	WB	-18.9	-12.8	-3.3	5.1	12.4	18.5	21.3	19.8	14.3	4.9	-5.7	-16.6		
		MCDB	-18.4	-11.6	-0.5	10.0	18.7	23.3	25.4	22.9	17.0	8.5	-4.3	-16.1		
(35) Mean Daily Temperature Range	MDBR		16.4	19.5	19.4	14.3	15.7	15.1	13.8	12.9	13.7	13.9	14.5	14.5		
	5% DB	MCDBR	16.8	20.3	19.7	18.5	20.9	19.1	17.0	15.8	16.7	18.6	14.2	15.8		
		MCWBR	16.0	18.4	15.3	11.5	11.8	11.2	9.8	10.6	11.2	12.8	12.3	15.0		
	5% WB	MCDBR	16.8	19.8	18.6	16.8	18.4	15.9	14.6	13.6	13.6	16.4	13.7	15.8		
		MCWBR	16.0	18.0	14.7	11.0	11.3	10.4	9.9	10.2	10.2	12.0	12.0	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.262	0.297	0.366	0.537	0.587	0.493	0.473	0.419	0.395	0.417	0.328	0.265		
	taud		2.134	2.055	1.913	1.692	1.690	2.023	2.134	2.283	2.251	1.999	2.066	2.090		
	Ebn at Noon		742	808	804	698	687	771	781	809	781	643	624	658		
	Edh at Noon		74	107	152	216	227	164	145	118	109	115	81	66		
(44) All-Sky Solar Radiation	RadAvg		1.56	2.76	4.37	5.05	5.12	5.63	5.23	4.60	3.53	2.29	1.54	1.21		
	RadStd		0.05	0.07	0.21	0.47	0.46	0.70	0.41	0.32	0.33	0.16	0.09	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.40	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page